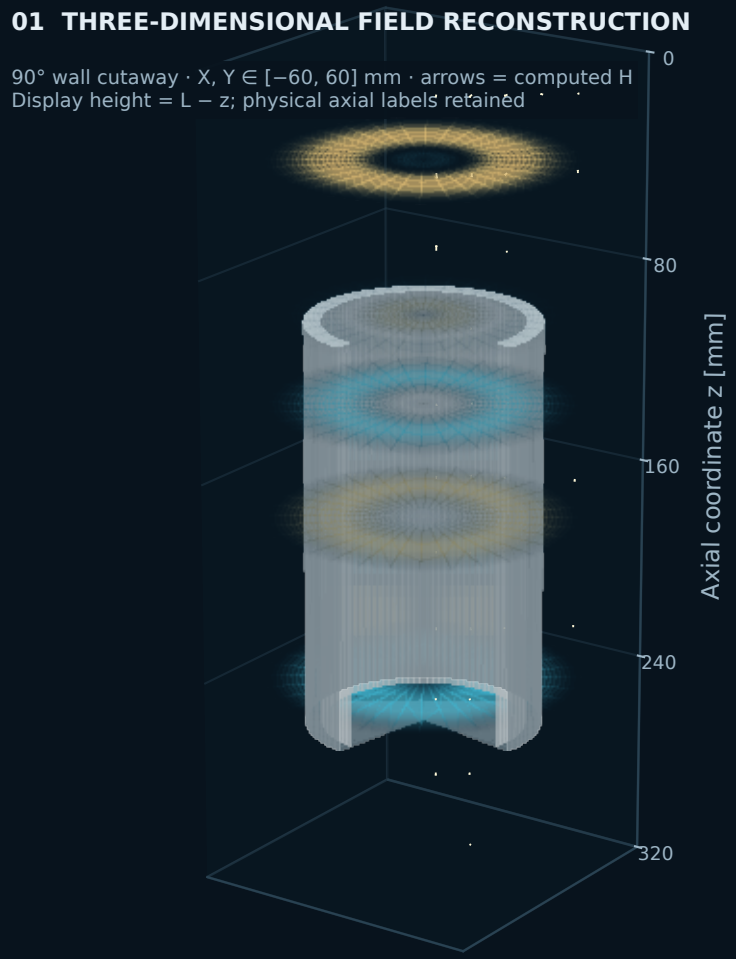
# GFM / CYLINDRICAL PULSE LABORATORY

**t = 1.840 ns**

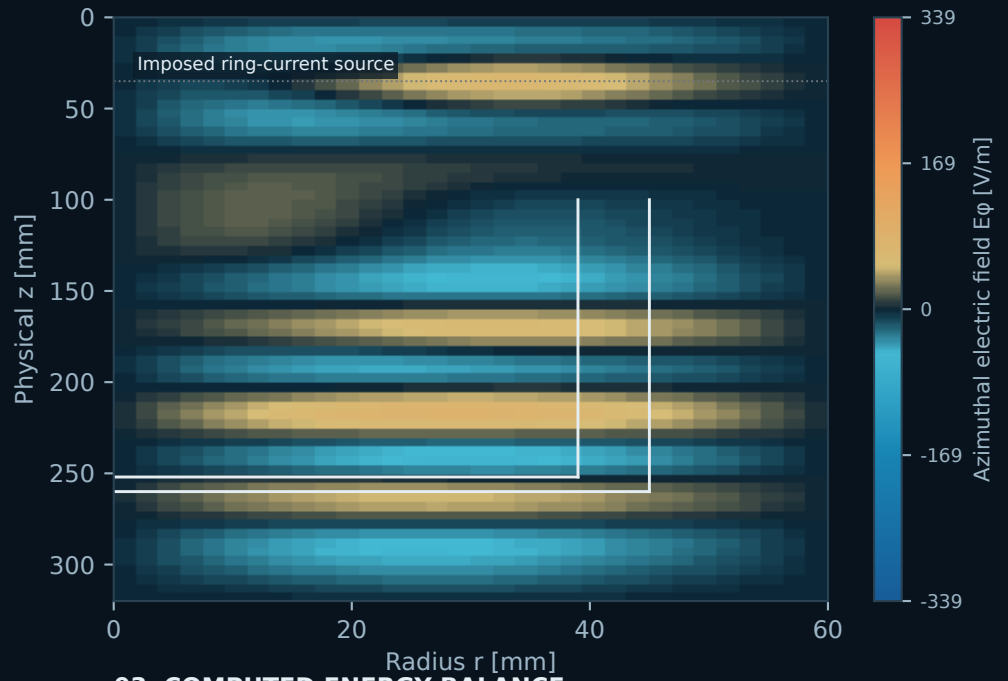
Calculated pulse transport into a hollow Ohmic cylinder · axisymmetric Maxwell reconstruction

## 01 THREE-DIMENSIONAL FIELD RECONSTRUCTION

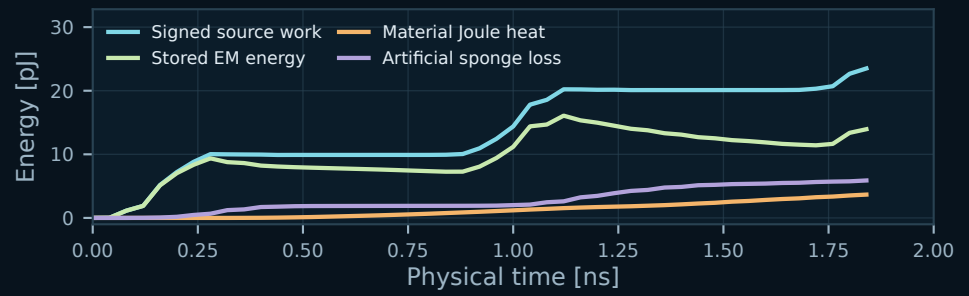



Hollow target Ø90 × 160 mm | wall 6 mm | σ = 0.02 S/m

## 02 EXACT r–z ELECTRIC SAMPLES




## 03 COMPUTED ENERGY BALANCE




## FIELD / MATERIAL RECORD

Eφ,max 83.306 V/m | Bmax 173.647 nT | deposited heat 3.6663 pJ

Peak matter ΔT 2.144e-14 K | relative energy residual 3.04e-14

Grid 32 × 64 annular cells | Δr = 1.875 mm Δz = 5.000 mm | cover 0%

Calculated m=0 electromagnetic fields; the displayed volume is a rotational reconstruction of the r–z grid.

Field colors are quantitative false color, without lighting. Gray surfaces indicate material geometry; color is not thermal emission.

PEC boundaries + electric sponge; constant vacuum ε, μ and generic Ohmic matter. No fluid, plasma dispersion or gravity evolution.

## EQUATIONS / DISPLAY REFERENCE

$\varepsilon\,\partial_t E_\varphi = \partial_z H_r - \partial_r H_z - \sigma E_\varphi + F_\varphi$

$\mu\,\partial_t H_r = \partial_z E_\varphi \qquad \mu\,\partial_t H_z = -r^{-1}\partial_r(rE_\varphi)$

Fixed color scale: $\pm E_{ref} = \pm F_0\tau / \varepsilon_0 = \pm 338.8$ V/m

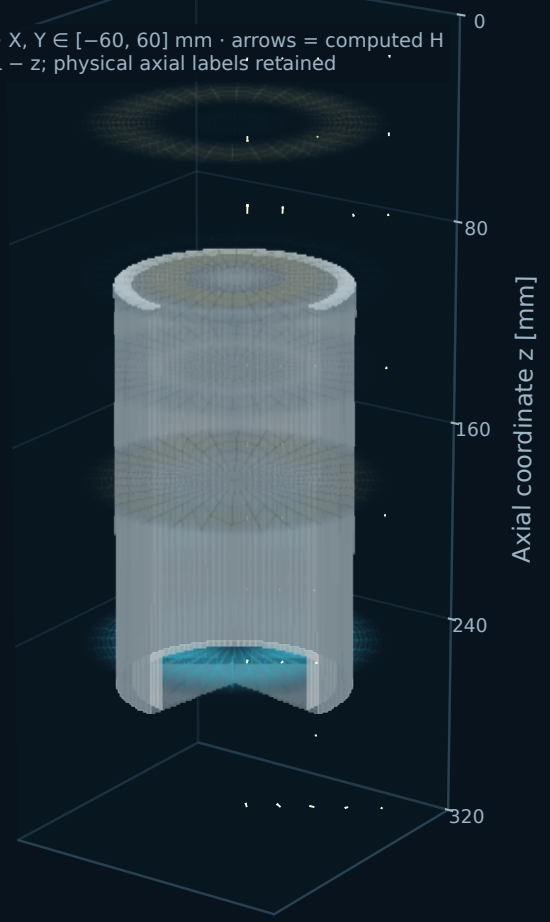
# GFM / CYLINDRICAL PULSE LABORATORY

**t = 2.000 ns**

Calculated pulse transport into a hollow Ohmic cylinder · axisymmetric Maxwell reconstruction

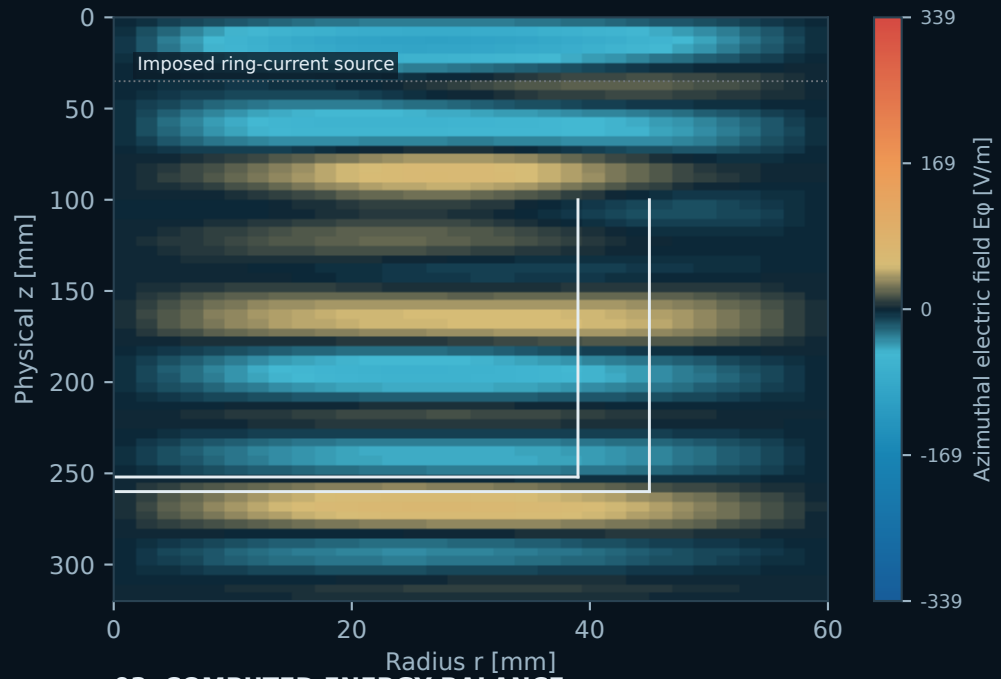
## 01 THREE-DIMENSIONAL FIELD RECONSTRUCTION

90° wall cutaway · X, Y ∈ [−60, 60] mm · arrows = computed H
Display height = L − z; physical axial labels retained

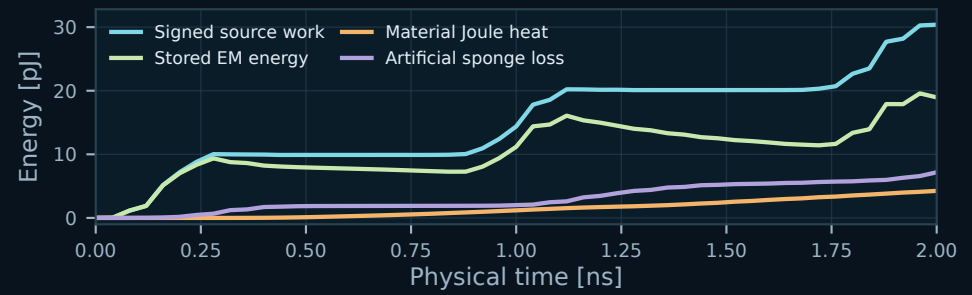



Hollow target Ø90 × 160 mm | wall 6 mm | σ = 0.02 S/m

## 02 EXACT r–z ELECTRIC SAMPLES




## 03 COMPUTED ENERGY BALANCE




## FIELD / MATERIAL RECORD

Eφ,max 103.261 V/m | Bmax 268.290 nT | deposited heat 4.2529 pJ
Peak matter ΔT 2.425e-14 K | relative energy residual 5.68e-14
Grid 32 × 64 annular cells | Δr = 1.875 mm Δz = 5.000 mm | cover 0%

Calculated m=0 electromagnetic fields; the displayed volume is a rotational reconstruction of the r–z grid.
Field colors are quantitative false color, without lighting. Gray surfaces indicate material geometry; color is not thermal emission.
PEC boundaries + electric sponge; constant vacuum ε, μ and generic Ohmic matter. No fluid, plasma dispersion or gravity evolution.

## EQUATIONS / DISPLAY REFERENCE

$\varepsilon\,\partial_t E_\varphi = \partial_z H_r - \partial_r H_z - \sigma E_\varphi + F_\varphi$

$\mu\,\partial_t H_r = \partial_z E_\varphi \qquad \mu\,\partial_t H_z = -r^{-1}\partial_r(rE_\varphi)$

Fixed color scale: $\pm E_{ref} = \pm F_0\tau / \varepsilon_0 = \pm 338.8$ V/m